

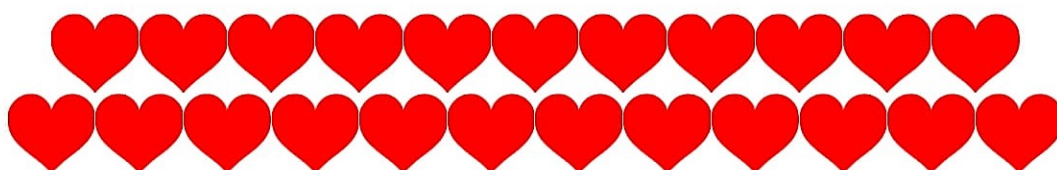
Lab 04

CS2

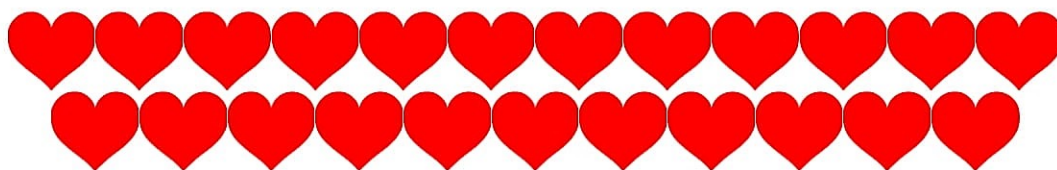
Instructor:

Dr Victor Govindaswamy

(100 points)



**Why did the deck of cards go to school?
To learn how to deal with life!**



Cultivating Today, the Innovators of Tomorrow

Educating Today, the Geniuses of Tomorrow

#wgp



Card Game Analysis App Called PokerPal

Objective:

This assignment reinforces concepts of pointers, 2D arrays, enums, and randomization. You will combine card shuffling with card evaluation to create a more functional program. By the end, you will have a program that simulates shuffling a deck of cards and evaluates hands for poker combinations.

You are also required to create a UML diagram.



Requirements

1. Single Class Implementation:

- Implement the entire program using a single class, CardGame.
- The class should encapsulate all data and functionality related to the deck, shuffling, dealing, and hand evaluation.

2. Menu-Driven Interface:

- Create a menu-based interface that allows the user to:
 1. Shuffle the deck.
 2. Deal a five-card poker hand.
 3. Display the current deck.
 4. Display the current hand.
 5. Evaluate the hand for poker combinations.
 6. Exit the program.

3. Deck Initialization:

- Represent a standard 52-card deck as a 2D array (4 rows for suits, 13 columns for ranks).
- Use an enum for suits (Spades, Hearts, Diamonds, Clubs) and ranks (Ace, 2, ..., King).

4. **Shuffling the Deck:**

- Implement a high-performance shuffling algorithm that iterates through the deck row by row and column by column. Swap each card with a randomly selected card in the deck.
- Display the shuffled deck to verify randomness.

5. **Dealing a Hand:**

- Implement a function to deal a five-card poker hand. Store the hand as a 2D array.

6. **Hand Evaluation:** Write member functions to evaluate the hand for the following poker combinations:

- Pair: Two cards of the same rank.
- Two Pairs: Two different pairs of cards.
- Three of a Kind: Three cards of the same rank.
- Four of a Kind: Four cards of the same rank.
- Flush: All cards of the same suit.
- Straight: Five cards of consecutive ranks.

Each function should return a boolean indicating whether the combination exists.

7. **User Interaction:**

- Provide a clear menu to guide the user through shuffling, dealing, displaying, and evaluating cards.
- After each hand is dealt, display the cards and the evaluation results (e.g., "Flush: Yes, Pair: No").



Specifications

- **Class Design:**
 - The class CardGame should contain:
 - A private 2D array for the deck.
 - A private 2D array for the current hand.
 - Public methods for initialization, shuffling, dealing, evaluating hands, and menu-driven interaction.
- **Input/Output:**
 - Display the shuffled deck as a grid (4 rows x 13 columns).
 - For each dealt hand, display the cards in a readable format (e.g., "Ace of Spades, 10 of Hearts").
 - Output the evaluation results for the poker combinations.
- **Functions to Implement:**
 - void initializeDeck();
 - void shuffleDeck();
 - void dealHand();
 - void displayDeck();
 - void displayHand();
 - bool isPair();
 - bool isTwoPair();
 - bool isThreeOfAKind();
 - bool isFourOfAKind();
 - bool isFlush();
 - bool isStraight();
 - void displayMenu();
 - void handleMenuOption(int option);
- **Constraints:**
 - Use a single class to encapsulate all functionality.
 - Focus on implementing the logic with 2D arrays and pointers.

- Handle dealt cards by marking them as “used” to avoid duplication.



Sample Output

```
Menu:
1. Shuffle Deck
2. Deal Hand
3. Display Hand
4. Evaluate Hand
5. Exit
Enter your choice: 1
Deck shuffled successfully!

Menu:
1. Shuffle Deck
2. Deal Hand
3. Display Hand
4. Evaluate Hand
5. Exit
Enter your choice: 2

Menu:
1. Shuffle Deck
2. Deal Hand
3. Display Hand
4. Evaluate Hand
5. Exit
Enter your choice: 3
Hand: 5 of Hearts, King of Diamonds, Ace of Spades, 10 of Clubs, 7 of Hearts
```

```
Menu:
1. Shuffle Deck
2. Deal Hand
3. Display Hand
4. Evaluate Hand
5. Exit
Enter your choice: 4
Evaluation:
Pair: No
Two Pair: No
Three of a Kind: No
Four of a Kind: No
Flush: No
Straight: No

Menu:
1. Shuffle Deck
2. Deal Hand
3. Display Hand
4. Evaluate Hand
5. Exit
Enter your choice: 2

Menu:
1. Shuffle Deck
2. Deal Hand
3. Display Hand
4. Evaluate Hand
5. Exit
Enter your choice: 3
Hand: Ace of Diamonds, Ace of Clubs, 5 of Spades, 8 of Hearts, Ace of Hearts
```

```
Menu:
1. Shuffle Deck
2. Deal Hand
3. Display Hand
4. Evaluate Hand
5. Exit
Enter your choice: 4
Evaluation:
Pair: Yes
Two Pair: No
Three of a Kind: Yes
Four of a Kind: No
Flush: No
Straight: No

Menu:
1. Shuffle Deck
2. Deal Hand
3. Display Hand
4. Evaluate Hand
5. Exit
Enter your choice: 5
Exiting the game.
```



Submission Requirements

- Submit a .h and cpp files containing your source code.
- Include comments to explain the purpose of each function and major sections of your code.
- Ensure the program compiles and runs without errors.



Follow-Up Questions:

1. How does the shuffling algorithm ensure fairness in randomization?
2. What challenges did you face while implementing poker hand evaluations?



Grading Criteria:

Criterion	Points
Correctness of Deck Initialization	10
Implementation of Shuffling Algorithm	15
Functionality of Menu Options	15
Accuracy of Poker Hand Evaluation	30
Code Organization and Comments	20
Compilation and Runtime Performance	10
Total	100



Additional Requirements:

- Validation: Ensure that you handle cases where there are no available slots for booking or no booked slots to cancel.
- Error Handling: If a user tries to cancel an appointment that doesn't exist, display an appropriate error message.
- Output: The system should display the schedule with clear formatting, showing booked and available slots for the doctor.



Happy Coding!